2021 Prelim Exams Paper 3 Answers

Qn	Answers	Mark
1a	Electric field strength is the electric force per unit positive charge	B1
1bi	$E = \frac{\Delta V}{d} = \frac{V_{higher} - V_{lower}}{d}$ Since electric field is upwards, $\therefore V_{lower} = +140\ V$ $1.40 \times 10^4 = \frac{V_{higher} - 140}{0.015}$ $V_{higher} = 350\ V$	 C1 A1
1bii1	$\mathrm{F_{net} = ma}$ $\mathrm{qE = ma_y}$ $(1.60 \times 10^{-19})(1.40 \times 10^4) = (9.11 \times 10^{-31})\mathrm{a_y}$ $\mathrm{a_y} = 2.4588 \times 10^{15} \approx 2.46 \times 10^{15}\ \mathrm{m\ s^{-2}}$ the direction of acceleration is downwards (towards the plate with higher potential) (ECF from 1bi)	 C1 A1 B1
1bii2	$s_x = u_x t + \frac{1}{2} a_x t^2$ $0.12 = (5.0 \times 10^7)t + 0$ $t = 2.4 \times 10^{-9}\ s$	 C1 A1
1bii3	Distance between mid point and bottom plate = $\frac{1.5}{2} = 0.75\ cm$ $s_y = u_y t + \frac{1}{2} a_y t^2$ $s_y = 0 + \frac{1}{2}(2.4588 \times 10^{15})(2.4 \times 10^{-9})^2$ $s_y = 7.08 \times 10^{-3} = 0.708\ cm < 0.75\ cm$ the electron does not hit the plates, thus passing between the plates. Or To hit the plate, the minimum vertical displacement = 0.75 cm $s_y = u_y t + \frac{1}{2} a_y t^2$ $0.0075 = 0 + \frac{1}{2}(2.4588 \times 10^{15})\mathrm{t}^2$ $t = 2.47 \times 10^{-9}\ s > 2.4 \times 10^{9}\ s$ the electron does not hit the plates, thus passing between the plates.	 C1 M1 A1